

APS110 – Lecture 21

See hand written notes for drawings.

Polymers, recall...

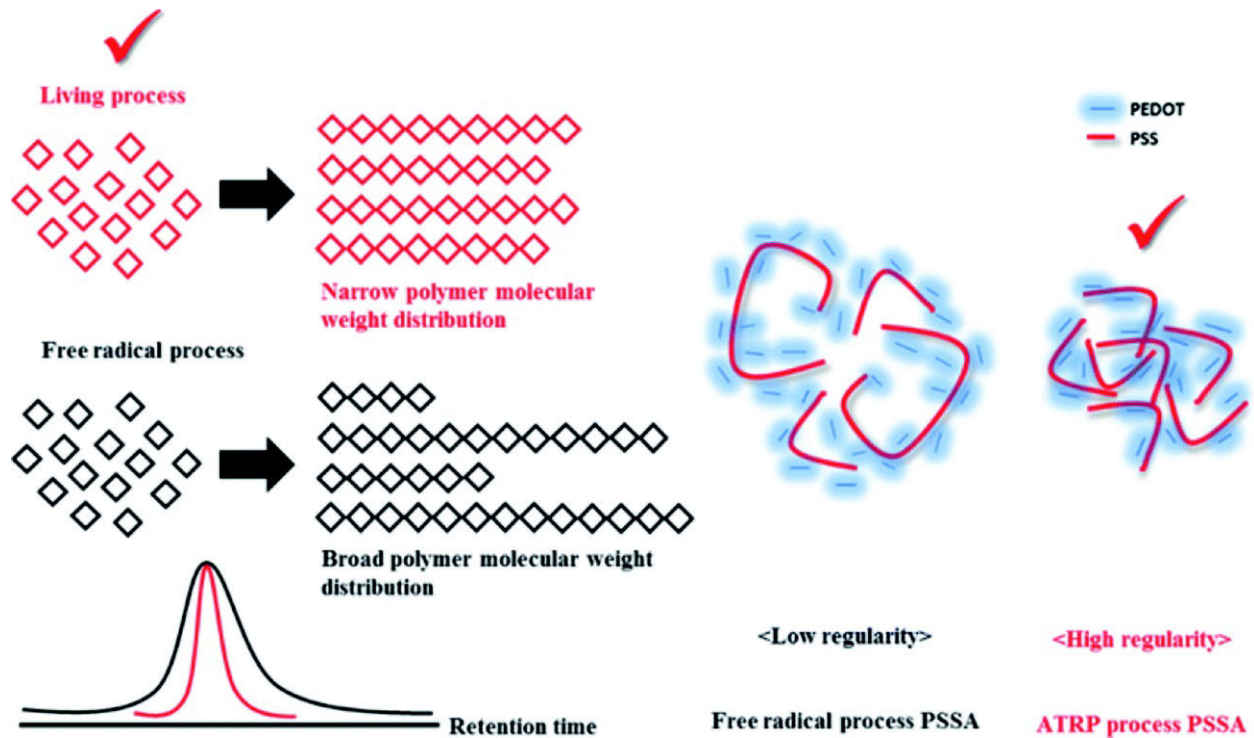
- Have lots of behaviours that are different from metals and ceramics → because of different microstructure
- Polymers are sensitive to how quickly you load them, they're sensitive to temperature changes

Molecular weight (g/mol) → most often used to quantify chain length

- When you think about molecular weight with polymers, DON'T think "how heavy?", think "**how long?**"
- If the molecules are really short, there isn't a lot of forces that are resisting pulling the molecules away from each other
- When the molecules are really long...
 - There is **more entanglement** → *physically* inhibited from pulling apart from other molecules
 - One of the mechanisms of strengthening
 - There are more weak bonds along the length of molecules
- Ex. noodles → a good analogy for polymer behaviour
 - Difficult to separate one noodle from the rest
 - If you cut the noodles, it makes it really easy to separate some noodles from the rest (reduce chain length!)

Why do we use molecular weight to quantify chain length?

- The longer the molecules, the more weight per mole of molecules
- Ex. UHMW PE (ultra-high-molecular-weight polyethylene)
 - Polymer used in car bumpers, docks, biomaterial in hip implants
 - Really carefully processed so the molecular weight of the polymer molecules is ultra-high → very very strong
 - High strength, high wear-resistance
- **Not all molecules in a sample will be the same length**
- Sometimes molecules can join together to form a really long molecule
- Sometimes a molecule, as it's building from mer units, the molecule will hit an imperfection or something and will stop getting longer (thus will be shorter)
- Careful process control can tighten the distribution down, but **there will always be a distribution of molecular weight (chain length)**
 - There can be a tight distribution or there could be a broad distribution
 - This distribution can shift to a direction
 - How will this affect the properties of the sample?
 - Wide distribution → in general, strength/service temperature/resistance to dissolution are DECREASED



How to quantify molecular weight?

- We use AVERAGE molecular weight
- But, a single average molecular weight doesn't describe the distribution...
 - You can have two samples with the same average but different distribution → different physical properties
- Different ways to calculate average MW
 - Number average (common way) → sum up all masses and divide by the number of molecules
 - Weight average → takes how much MW there is contributed by each polymer to the total weight
 - Not as commonly used, but it is important in polymer chemistry
 - You can compare the number average and the weight average
 - The only time the number average and weight average would be the same is if the distribution was really tight → this is one way we can use to **quantify the distribution**

Another way to increase strength → introduce **CROSS-LINKS**

- Important!! → doesn't just increase strength, it increases modulus, and resistance to plastic deformation
- Introduce bonds between the chains of polymers
- Limit relative movement of molecules
 - This is of course the basis of plastic deformation, and so the cross-links strengthen the sample
- These bonds can be either STRONG primary bonds or WEAK secondary bonds

- o Introducing strong bonds is the primary way of strengthening with cross-linking
- The pitfall of cross-links is that the polymers will not melt
 - o Melt-able polymers are easy to work with (melting, molding, re-melting, etc.)
 - o There is research going into making thermoplastics that are also elastomers
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